Job Salary Prediction Dataset

Exploratory Data Analysis & Hypothesis Testing — Student Assignment

Dataset Overview

File: job_salary_prediction_dataset.csv

Rows: 250,000 | Columns: 10 (9 features + 1 target)

Target variable: salary (continuous, numeric)

Column	Type	Description
job_title	Categorical	Job role (12 categories, e.g. Data Scientist, DevOps Engineer)
experience_years	Numeric (discrete)	Years of professional experience
education_level	Categorical (ordinal)	High School, Diploma, Bachelor, Master, PhD
skills_count	Numeric (discrete)	Number of professional skills listed
industry	Categorical	Industry sector (10 categories)
company_size	Categorical (ordinal)	Startup, Small, Medium, Large, Enterprise
location	Categorical	Country / region of employment (10 categories, incl. Remote)
remote_work	Categorical	No, Hybrid, Yes
certifications	Numeric (discrete)	Number of certifications held
salary	Numeric (continuous) — TARGET	Annual salary

Instructions

- Use Python (pandas, matplotlib, seaborn, plotly, scipy, statsmodels, or equivalent libraries of your choice).
- For every visualization, choose the chart type yourself — no chart type is specified for you — and briefly justify why that chart type is appropriate for the variable(s) involved.
- For every statistical test in Part B, you must: (i) state the null and alternative hypotheses, (ii) check any relevant assumptions of the test, (iii) select and run an appropriate test, and (iv) justify in your own words why that specific test — and not an alternative — was the correct choice given the data types and assumption checks.
- Report and interpret p-values, test statistics, and effect sizes (where applicable) at a 0.05 significance level unless stated otherwise.
- Include your code, output, and written interpretation for every question.

Part A — Exploratory Data Analysis (Visualization)

Answer the following using appropriate visualizations. For each answer, include the plot, the code used to generate it, and 2–4 sentences interpreting what the plot shows.

A1. Univariate Exploration

1. Examine the distribution of the target variable, salary. Is it symmetric, skewed, or does it show any unusual patterns (e.g. multiple peaks, gaps, outliers)?

2. Examine the distribution of experience_years. How are experience levels spread across the workforce represented in this dataset?

3. Examine the distributions of skills_count and certifications. Do most employees cluster around particular values, or are these fairly evenly spread?

4. For each categorical column (job_title, education_level, industry, company_size, location, remote_work), visualize how many records fall into each category. Which categories are over- or under-represented?

A2. Bivariate Exploration — Target vs. Numeric Features

5. Visualize the relationship between experience_years and salary. Describe the shape of the relationship you observe (linear, curved, none, etc.) and whether it seems consistent across the full range of experience.

6. Visualize the relationship between skills_count and salary, and separately between certifications and salary. Do either of these appear to matter as much as experience?

7. Produce a single visualization that summarizes the pairwise relationships among all numeric variables (including the target) at once. What does it suggest about which numeric feature is most strongly associated with salary?

A3. Bivariate Exploration — Target vs. Categorical Features

8. Compare the salary distribution across the different education_level categories. Does salary appear to increase consistently with education level, or is the pattern less clear?

9. Compare the salary distribution across company_size categories and separately across remote_work categories. Summarize any noticeable differences.

10. Compare the salary distribution across job_title and separately across industry. Which job titles or industries stand out as paying noticeably more or less, and are there wide spreads within any single category?

11. Compare the salary distribution across location. Are there visible differences by country/region, and how would you characterize the spread within each?

A4. Multivariate & Outlier Exploration

12. Create a visualization that shows how salary varies jointly across two categorical variables at once (e.g. combining education_level with remote_work, or company_size with industry — your choice of pairing). What combined pattern, if any, emerges that isn't obvious from either variable alone?

13. Identify potential outliers in salary and in at least one numeric feature using an appropriate visualization. How many extreme values do you find, and would you treat them as data errors or as legitimate high/low values?

14. Choose one additional visualization not covered above that you believe reveals something useful about this dataset. Justify your choice of variables and chart type.

Part B — Testing the Relationship Between Target and Features

For each question below, formally test whether a statistically significant relationship exists between salary and the specified feature(s). You must select the test yourself based on the variable types involved and the assumptions the data satisfy (or violate).

B1. Numeric Features vs. Salary

1. Test whether there is a statistically significant relationship between experience_years and salary. Check whatever assumptions are relevant to the test you choose before running it, and justify your final choice of test based on those checks.

2. Test whether there is a statistically significant relationship between skills_count and salary.

3. Test whether there is a statistically significant relationship between certifications and salary.

> *For each of the above, report the test statistic, p-value, and a measure of the strength of the relationship — and justify why that strength measure is appropriate.*

B2. Categorical Features vs. Salary

4. Test whether mean or median salary differs significantly across education_level groups.

5. Test whether mean or median salary differs significantly across company_size groups.

6. Test whether mean or median salary differs significantly across remote_work groups.

7. Test whether mean or median salary differs significantly across job_title groups.

8. Test whether mean or median salary differs significantly across industry groups.

9. Test whether mean or median salary differs significantly across location groups.

> *For each of the above, justify your choice of test by explicitly checking and reporting on the relevant assumptions (e.g. normality within groups, equality of variances, sample size per group). If your chosen test indicates a significant overall difference, follow up to identify which specific group(s) differ.*

B3. Combined / Model-Based Relationship Testing

10. Fit a model that uses all nine features together to explain salary. Test whether the overall model is statistically significant, and test the significance of each individual feature's contribution. Justify your choice of modeling approach given the mix of numeric and categorical predictors, and check whatever assumptions your approach requires.

11. Based on everything tested in Part B, rank the features from strongest to weakest relationship with salary. Justify your ranking — including why you chose the particular statistic(s) you used to compare relationships that were tested using different methods.

Submission note: Submit a single notebook or report containing all code, plots, test outputs, and written justifications, organized by question number.